Objective

To make sure the data is clean, correct, and ready for analysis.

Categorical Data Cleaning

- Standardized categorical columns like **business_unit** and **account_type** using Power Query
- Fixed different spellings, extra spaces, and text case
- Made all values consistent (Proper Case)

Result:
Categories are uniform and easy to group in analysis.

Date Validation

- Checked that all **transaction_date** values are in correct date format
- Invalid or missing dates were flagged for review

Result:
Dates can be safely used for monthly and yearly analysis.

Missing Values Check

- Checked important columns:
    - transaction_date
    - amount
    - account_type
    - business_unit
- Missing values were **identified and flagged**, not deleted

Result:
Rows with missing data are clearly marked for review.

Outlier Detection (Amount)

- Used the **IQR method** to find unusually high or low transaction amounts
- Upper and lower limits were calculated
- Amounts outside this range were flagged as outliers

Result:
No extreme outliers were found in the dataset.

Validation Status

- Created a **validation_status** column
- Combined missing value and outlier checks into one final flag:
    - **Valid** → Data is ready for analysis
    - **Review** → Data needs checking

Result:
Most records are valid and suitable for analysis.

Final Conclusion

The dataset is mostly clean and reliable. All data quality issues were flagged without deleting any records. The cleaned and validated data is ready for further analysis.